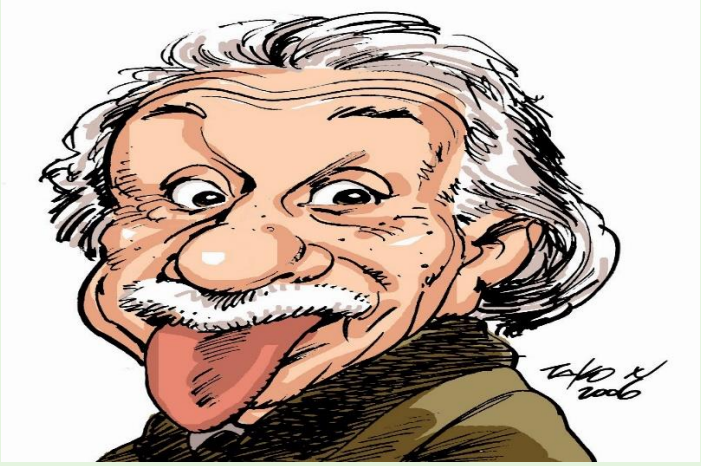
EXERCÍCIO 1

Aluno: Felipe Moreira Rios

CURSO: Engenharia de Computação

Quesito 1

Identifique o gênero textual representado nas figuras abaixo.



A - Caricatura

WEAR A MASK

AND SAVE YOUR LIFE!

The Emergency That Now Confronts Our City Is Beyond the Facilities of the Health Department

The RED CROSS

has come to the assistance of the Board of Health. Doctors and nurses can not be obtained to take care of the afflicted. You must wear a mask, not only to protect yourself but your children and your neighbor from influenza, pneumonia and death

A GAUZE MASK IS 99% PROOF AGAINST INFLUENZA

Doctors wear them. Those who do not wear them get sick. The man or woman or child who will not wear a mask now is a dangerous slacker.

WEAR MASKS

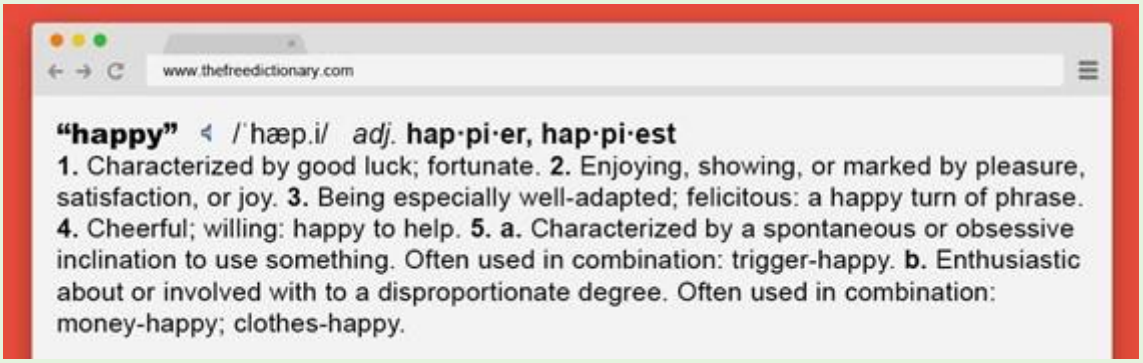
This statement was authorized at a meeting of the undersigned, who are convinced that it is the only way to stamp out the epidemic. You must do your part

Alameda County Relief Committee

County of Alameda — Oakland Chapter American Red Cross — Associated Charities
City of Oakland — Oakland Clearing House Association — Retail Dry Goods Association
Board of Health of Oakland — Oakland Chamber of Commerce

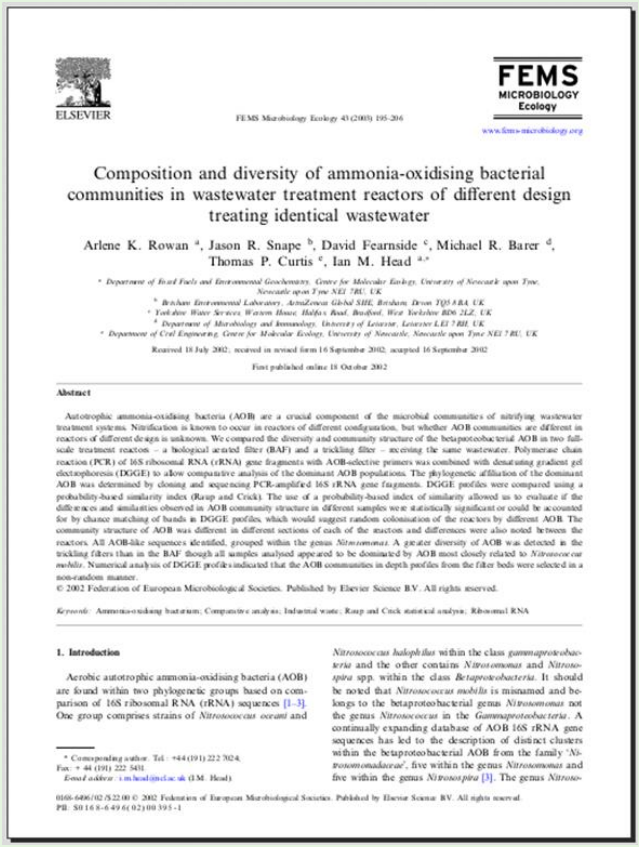
B – Propaganda jornalística

C – Mensagem de Texto



D - Verbete

E - Charge



ELSEVIER

FEMS Microbiology Ecology 43 (2003) 195–206

FEMS MICROBIOLOGY Ecology

www.fems-microbiology.org

Composition and diversity of ammonia-oxidising bacterial communities in wastewater treatment reactors of different design treating identical wastewater

Arlene K. Rowan [a], Jason R. Snape [b], David Fearnside [c], Michael R. Barer [d], Thomas P. Curtis [e], Ian M. Head [a,*]

[a] Department of Fossil Fuels and Environmental Geochemistry, Centre for Molecular Ecology, University of Newcastle upon Tyne, Newcastle upon Tyne NE1 7RU, UK
[b] Brixham Environmental Laboratory, AstraZeneca Global SHE, Brixham, Devon TQ5 8BA, UK
[c] Yorkshire Water Services Western House, Halifax Road, Bradford, West Yorkshire BD6 2LZ, UK
[d] Department of Microbiology and Immunology, University of Leicester, Leicester LE1 7RH, UK
[e] Department of Civil Engineering, Centre for Molecular Ecology, University of Newcastle, Newcastle upon Tyne NE1 7RU, UK

Received 18 July 2002; received in revised form 16 September 2002; accepted 16 September 2002

First published online 18 October 2002

Abstract

Autotrophic ammonia-oxidising bacteria (AOB) are a crucial component of the microbial communities of nitrifying wastewater treatment systems. Nitrification is known to occur in reactors of different configuration, but whether AOB communities are different in reactors of different design is unknown. We compared the diversity and community structure of the betaproteobacterial AOB in two full-scale treatment reactors – a biological aerated filter (BAF) and a trickling filter – receiving the same wastewater. Polymerase chain reaction (PCR) of 16S ribosomal RNA (rRNA) gene fragments with AOB-selective primers was combined with denaturing gradient gel electrophoresis (DGGE) to allow comparative analysis of the dominant AOB populations. The phylogenetic affiliation of the dominant AOB was determined by cloning and sequencing PCR-amplified 16S rRNA gene fragments. DGGE profiles were compared using a probability-based similarity index (Raup and Crick). The use of a probability-based index of similarity allowed us to evaluate if the differences and similarities observed in AOB community structure in different samples were statistically significant or could be accounted for by chance matching of bands in DGGE profiles, which would suggest random colonisation of the reactors by different AOB. The community structure of AOB was different in different sections of each of the reactors and differences were also noted between the reactors. All AOB-like sequences identified, grouped within the genus *Nitrosomonas*. A greater diversity of AOB was detected in the trickling filters than in the BAF though all samples analysed appeared to be dominated by AOB most closely related to *Nitrosococcus mobilis*. Numerical analysis of DGGE profiles indicated that the AOB communities in depth profiles from the filter beds were selected in a non-random manner.

© 2002 Federation of European Microbiological Societies. Published by Elsevier Science B.V. All rights reserved.

Keywords: Ammonia-oxidising bacterium; Comparative analysis; Industrial waste; Raup and Crick statistical analysis; Ribosomal RNA

1. Introduction

Aerobic autotrophic ammonia-oxidising bacteria (AOB) are found within two phylogenetic groups based on comparison of 16S ribosomal RNA (rRNA) sequences [1–3]. One group comprises strains of *Nitrosococcus oceani* and *Nitrosococcus halophilus* within the class *gammaproteobacteria* and the other contains *Nitrosomonas* and *Nitrosospira* spp. within the class *Betaproteobacteria*. It should be noted that *Nitrosococcus mobilis* is misnamed and belongs to the betaproteobacterial genus *Nitrosomonas* not the genus *Nitrosococcus* in the *Gammaproteobacteria*. A continually expanding database of AOB 16S rRNA gene sequences has led to the description of distinct clusters within the betaproteobacterial AOB from the family '*Nitrosomonadaceae*', five within the genus *Nitrosomonas* and five within the genus *Nitrosospira* [3]. The genus *Nitroso-*

* Corresponding author. Tel.: +44 (191) 222 7024; Fax: +44 (191) 222 5431.
E-mail address: i.m.head@ncl.ac.uk (I.M. Head).

0168-6496/02/$22.00 © 2002 Federation of European Microbiological Societies. Published by Elsevier Science B.V. All rights reserved.
PII: S0168-6496(02)00395-1

F – Artigo científico

Quesito 2

Complete o texto com as cognatas disponíveis

i. VADER is applied to identify **suspicious** activities from texts posted on **social** media sites, particularly Facebook and Twitter, to classify sentiments into different **categories** of positive, negative, and neutral along with obtaining the **intensity** of emotions.

social categories suspicious intensity

ii. VADER, a lexicon based sentimental **analysis**, is utilized to **identify** the suspicious activities from texts **posted** on social media sites particularly Facebook and Twitter

posted identify analysis

iii. Accordingly, it becomes very **critical** to find out opinion sharing patterns and dig them to exploit the individuals' **suspicious** behaviors as a social media **platform** is a haven for these people to communicate with their companions.

platform critical suspicious

iv. Generally, it aims at **processing** data into such a form that can be easily used for **converting**. In this study, Figure 3 show an **example** of data preprocessing phase undergoes several steps involving.

processing example converting

Quesito 3

Leia o texto abaixo e marque com amarelo as palavras cognatas e com azul as palavras que você conhece de experiências prévias. Se você escreve seus exercícios, sublinhe as cognatas e circule as palavras de experiência prévia.

i. On average, accidents per flight hour of occur 100 times more often than with manned aircraft. The main factors of high accident rate are frequent assembly and disassembly of systems, ignoring collection and analysis of information about the nature of component failures, low level of service and personnel qualification.

ii. This study proposes the use of mathematical method of artificial neural networks for aircraft maintenance. Statistics on the operation of individual components will be used to neural networks learning, and the predicted reliability analysis based on operational and maintenance history data are used as a decision support system (DSS), to provide recommendations for detailed inspections and, if necessary, further repair or maintenance.

iii. The accuracy of the results is compared with the traditional method of mathematical interpolation. Also, the evaluation of the analysis of the nonlinear factors, the introduction of additional perturbations starts at a specific time. This simulates cumulative factors such as cosmic radiation. This work is a contribution to the promising UAV service models.